LEDGERLINE

Data Model

VERSION	1.0.0
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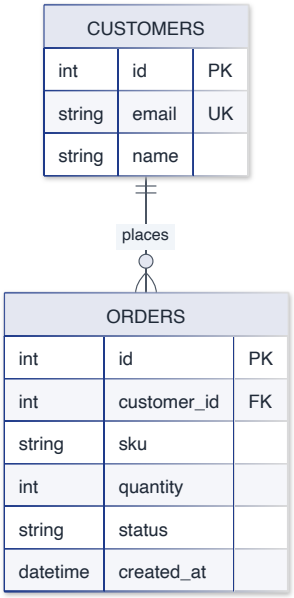
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Overview

Database	PostgreSQL 16 (`postgres:16` in `docker-compose.yml`)
Access	SQLAlchemy 2.0 ORM, typed `Mapped[...]` models
Source	`app/models.py` (models), `alembic/versions/` (migrations)
Entities	2: `Customer`, `Order`

Entity relationship diagram



Entities

Customer

A wholesale buyer (`app/models.py`, table `customers`).

Field	Type	Null	Key	Notes
`id`	int	no	PK	
`email`	varchar(255)	no	unique	
`name`	text	no		

Order

An order line for one SKU (`app/models.py`, table `orders`).

Field	Type	Null	Key	Notes
`id`	int	no	PK	

Field	Type	Null	Key	Notes
`customer_id`	int	no	FK → `customers.id`	indexed
`sku`	varchar(64)	no		
`quantity`	int	no		
`status`	text	no		default `pending`
`created_at`	timestamp	no		default `datetime.utcnow`

Relationships

From	To	Cardinality	On delete	Evidence
Customer	Order	1:N	database default ASSUMPTION	`relationship(back_populates=...)` in `app/models.py`

Indexes and constraints

Table	Index / constraint	Columns	Type
`customers`	primary key	`id`	btree
`customers`	unique	`email`	btree
`orders`	primary key	`id`	btree
`orders`	index	`customer_id`	btree
`orders`	foreign key	`customer_id` → `customers.id`	constraint

Migrations

- Tool: Alembic · Location: `alembic/versions/` · Count: 2
- `0001_init.py`: create customers and orders
- `0002_order_status.py`: add order status

```
make migrate     # alembic upgrade head
```

> **NOTE**
> `created_at` uses `datetime.utcnow`, which returns naive timestamps. Store and compare times as UTC throughout.

Related documents

API Reference · Architecture

Assumptions and open questions

#	Item	Type	Why unverified	Who can confirm
A1	On-delete behaviour for orders	assumption	no `ondelete=` set in the model	backend lead
Q1	Should `status` be an enum?	open question	free text today	backend lead